Understanding value through agent-based modeling

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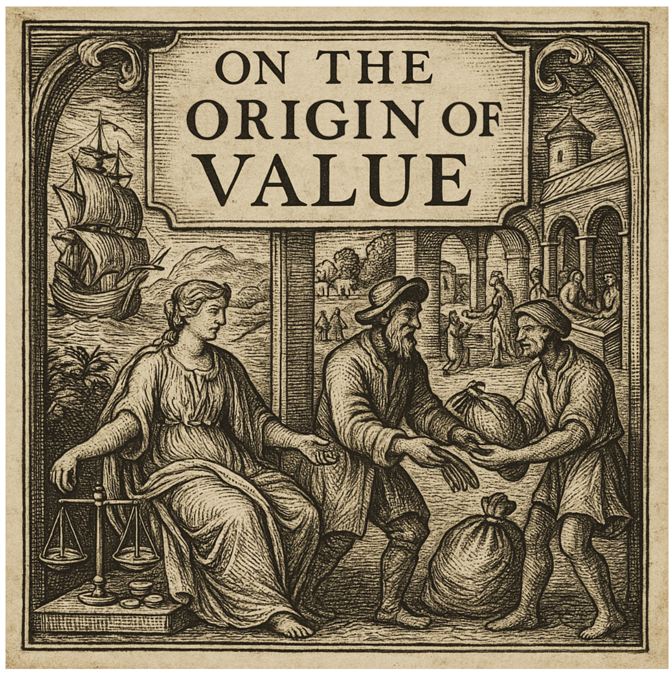

Abstract

This thesis builds upon the agent-based model developed by R. Donangelo and K. Sneppen to better understand how value emerges and behaves in decentralised trading systems. We examine three core regimes of the model - pure need, pure greed, and mixed trading - using both analytical and numerical methods. In the pure need regime, demand stabilises and fluctuates around direct need, with limited long-term dynamics ($H \approx 0.48$). In contrast, greed-driven trading leads to convergence toward a shared memory and randomness in demand ($H \approx 0.5$). The mixed regime reveals richer behaviour, where positive and negative feedback loops drive persistent demand fluctuations ($H \approx 0.82$), arising from structured need chains and emergent collective memory. Finally we attempt to incorporate elements of real world markets into the model, by letting agents differ in memory capability or starting wealth. While the model remains analytically limited in general cases, it offers a simple framework for exploring how basic trading rules can generate persistent demand dynamics similar to those seen in real markets.

Contents

Introduction

Value is a key point of analysis, given its fundamental importance from local to global trade. While driving some of the most consequential decisions in history, value has also been surrounded by a persistent dissensus, making it one of the most debated concepts among economists, philosophers, and politicians alike.

During classical economics, the Labour Theory of Value proposed that a product's worth was proportional to the labour required to produce it - if one product took twice as long to make, it was deemed twice as valuable [1]. In modern theory, price is governed by supply and demand, making Labour Theory essentially a supply-based view: goods that are harder to produce are scarcer, and thus more valuable. Although still referenced, most economists have since replaced it with the Subjective Theory of Value [1]. In other words, instead of viewing value as intrinsic, economists like Carl Menger argued that value is subjectively determined by individual needs - placing emphasis on demand rather than production:

Value is therefore nothing inherent in goods, no property of them, but merely the importance that we first attribute to the satisfaction of our needs — Principles of Economics, 1871 [2, p. 116]

Rather than declare one theory "right," econophysicists often build statistical-mechanics models to explore complex systems [3]. While modern economics locates price at the supply-demand intersection, the Subjective Theory argues that value ultimately arises from individual preferences (demand) and needs [2]. Throughout this paper, we focus primarily on the effect of demand on value, basing our analysis on the Subjective Theory of Value. We use and analyse a variation of the agent-based trading model by R. Donangelo and K. Sneppen [4], exploring the origin of value and the importance of the interplay between need and desire.

As a final step, we aim to relate our model to real-world markets by simulating trading and agent behaviour under conditions inspired by actual economic systems. Here we aim to explore how value emerges and evolves under varying circumstances - such as unequal wealth distributions or systems where agents have varying memory capacities.

Method

Having asked the question of how "value" can arise through trading - without any central price setter - we now turn to the precise model we will use to explore this phenomenon. In this chapter, we present our variation of the Donangelo-Sneppen model [4], which serves as our computational laboratory for studying demand driven value formation. We begin by outlining the core mechanics and parameters of the model - namely the trading rules, memory and inventory structures, and the system's initial state - so that we can understand exactly how the agents interact. Following this, we introduce key concepts and analytical metrics (such as demand, direct need, and the Hurst exponent) that allow us to measure and interpret market-like behaviour throughout the thesis.

Implementation of the trading model

Our model, based on the Donangelo-Sneppen model [4], is a simple agent-based trading model. The idea is to simulate a market where agents trade products with one another according to two different trading patterns: need-based and greed-based trading. The model is built on barter exchanges - that is, the exchange of goods without the use of money. Each agent possesses certain products and trades them with other agents, thereby simulating a barter-driven market.

A) Initial Inventory of Agents

Agent \ Product	1	2	3	4	5	6	7	8	...	L
1	2	1	0	2	0	1	0	1		2
2	0	0	2	1	1	1	3	0		1
⋮	|Amount held of each product type|									
N	2	1	2	0	1	2	0	0		1

B) Initial Memory List of Agents | L = 10 |

Agent \ Memory	1	2	3	4	5	6	7	8	...	M
1	1	9	9	4	6	5	1	2		1
2	8	5	7	7	6	2	8	7		10
⋮	|Index representing each product type|									
N	9	3	4	6	2	2	8	10		5

Figure 1: Initialisation of agent inventories and memories in the trading model. A) Each agent is assigned an initial inventory with counts of each product. To assure even wealth distribution the sum of each row adds up to U. B) Agents are also initialised with a memory list containing product indices, representing their individual preferences or awareness of different products.

Agent initialisation

The agent model consists of N agents that can trade L distinct products with each other. Each agent is initialised with U random copies of the L different products, ensuring only that, across all agents, there is an equal number of copies of each of the L products. To make the agents unique, we give each of them their own inventory, which tracks how many copies of each product they hold, as well as a memory that records past trade requests. At time t, $I_{i,j}(t) \in 0, 1, 2, \ldots$, is how many copies of product j agent i holds. Each agent has M memory slots, $T_{i,1}(t), \ldots, T_{i,M}(t)$, where each entry represents a product that this agent has been requested in a previous trade, thus taking values from $1, \ldots, L$. These two matrices are illustrated in Figure 1, at a random time t_0.

Trading procedure

At each time step of the simulation, we randomly pair the N agents, forming $N/2$ pairs. Each pair then attempts one barter exchange, resulting in $N/2$ exchange attempts per unit time. In each trade attempt, a requesting agent i choose a single product j to request from its partner. This choice follows a two-stage decision process:

1. **Need-Based Trading:** The default way for an agent to choose which product to request is based on need. If an agent is completely missing a product j, that is, $I_{i,j}(t) = 0$, we say that agent i has a need for product j. If the agent has any such needs, it will randomly choose one of these needed products to request in the subsequent trade. This form of requesting a product is denoted as Need-based trading.

2. **Greed-Based Trading:** Given that an agent has no needs - that is, $I_{i,j}(t) \geq 1$ for all $j = 1, \ldots, L$ - he defaults to requesting a product based on his own memory. In what we define as greed-based trading, the agent will request a product by randomly choosing one that appears in his memory $T_{i,\bullet}(t)$. Specifically, agent i will request product j with

probability

$$P(\text{request } j) = \frac{K_{i,j}(t)}{M},$$

where $K_{i,j}(t)$ is the number of times product j appears in agent i's memory $T_{i,\bullet}(t)$, and M is the total size of the memory.

Note that $P\big(\text{request } j\big)$ is identical to the probability used in the Donangelo-Sneppen model. In their formulation, however, it determines the chance that agent i would accept product j in a trade [4, p. 574]. In our model, trades are always accepted whenever possible. The only scenario in which a trade can fail is if agent i is requested a product j that it does not possess. Whether the barter succeeds or not, each agent i replaces a randomly chosen memory entry from $T_{i,\bullet}$ with the index of the good that their partner requested from them.

This setup, where agents act solely based on their own memory and inventory, closely reflects the idea proposed by Hayek in his famous work *The Use of Knowledge in Society* [5]: that economic order and coordination can emerge spontaneously from local interactions, without any central authority. To investigate whether such emergent order appears in our model, we now turn to a set of analytical metrics designed to capture both individual behaviour and market-level patterns.

Metrics for analysis

Before analysing the model from a theoretical standpoint, we will define some key parameters to describe the model's behaviour. These metrics are intended to reflect micro-level interactions as well as macro-level market dynamics, helping us trace how local exchanges contribute to broader patterns in value and performance. They also provide a basis for comparing different model regimes and evaluating the extent to which the system resembles real-world economic behaviour.

Demand

In a barter system like ours, there is no currency to represent the desirability of a product. To study how "value" emerges in the absence of a central price setter, we use *demand* as a proxy for monetary worth. We define demand $D(j)$ at time t as the fraction of all memory entries devoted to good j:

$$D(j) \ = \ \frac{1}{NM}\sum_{i=1}^{N} K_{i,j}(t), \tag{1}$$

where NM is equal to the total amount of memory slots in the system, functioning as a normalization factor. Demand is intended to reflect how desirable each product is to the agents, representing how highly they value each product. We interpret $D(j)$ as a form of "price," since a high value of $D(j)$ indicates that many agents want product j, suggesting that agents perceive it as having a higher exchange value.

Because all goods are intrinsically identical [2], a perfectly uniform market would have $D(j) \approx \frac{1}{L}$ for every j. In practice however, we will see that goods can accumulate higher $D(j)$.

Direct need

Need is defined as an agent having zero copies of a product. For analysis we also calculate the **direct need** for each individual item, defined as the percentage of the total need across all agents.

$$A(j) \;=\; \frac{a_j(t)}{\sum_{k=1}^{L} a_k(t)} \tag{2}$$

Here $a_j(t)$ is the number of agents in need of product j at time t. In other words, $A(j)$ represents how much a specific item is needed relative to the total number of unmet needs in the system. As such, it serves as a direct signal of scarcity, which - together with demand - helps us analyse how value emerges. By comparing $A(j)$ with $D(j)$, we can observe whether value aligns with actual need, as suggested my Menger [2], or if it is driven by other dynamics such as speculation or memory effects. Since $\sum_{j=1}^{L} a_j(t)$ is simply the total number of unmet needs across all agents, $A(j) \approx \frac{1}{L}$ if all missing items are uniformly distributed, but $A(j) \to 1$ if nearly every agent lacks j while having everything else.

Performance metric

To compare how effectively each agent is trading, we need a scoring system that allows us to evaluate different trading strategies and agent behaviours. Since our model assumes that all trades are accepted when possible, we cannot rely on trade frequency alone as a meaningful measure of performance. Instead, we define profit based on the difference in demand between the good an agent receives and the good it gives away. This approach reflects whether agents are trading up or down in terms of value, and aligns with the idea of using demand as a proxy for price.

$$\Delta x_i(t) \;=\; D(r, t-1) - D(f, t-1) \tag{3}$$

Here, r represents the good received, and f the good given away. $\Delta x_i(t)$ can be interpreted as the profit or loss resulting from bartering two products - similar to how a stockbroker might gain or lose money by exchanging one stock for another, depending on their relative prices. To prevent current trades from affecting prices, $\Delta x_i(t)$ is calculated using the demand values from the previous time step, $D(j, t-1)$. Because agents engage solely in barter exchanges, this scoring system is inherently zero-sum, one agent's gain is another's loss.

Hurst exponent

In our simulation, we aim to replicate real market behaviour as closely as possible, without overcomplicating the model. One important characteristic of financial markets - particularly stock markets - is their tendency to follow subtle trends that emerge from collective human behaviour. These trends are often too weak to be easily detected or exploited; but they can be captured statistically.

One such indicator is the Hurst exponent. The Hurst exponent, denoted H, is a value between 0 and 1 that characterises the behaviour of a time series, such as price fluctuations over time. If $H \approx 0.5$, the series behaves like a random walk - similar to flipping a coin. If $H < 0.5$, the series is mean-reverting, meaning it tends to reverse direction. If $H > 0.5$, the series exhibits persistence, meaning it tends to continue in the same direction (e.g., if prices have been rising, they're likely to keep rising, and vice versa).

For our model, we ideally want to capture a Hurst exponent in the interval $[0.55, 0.65]$, as this is the empirically observed range for most stock markets [6]. It is important to emphasise that the Hurst exponent reflects statistical tendencies over a broad range of time scales, rather than just short- or long-term trends or directly tradable signals.

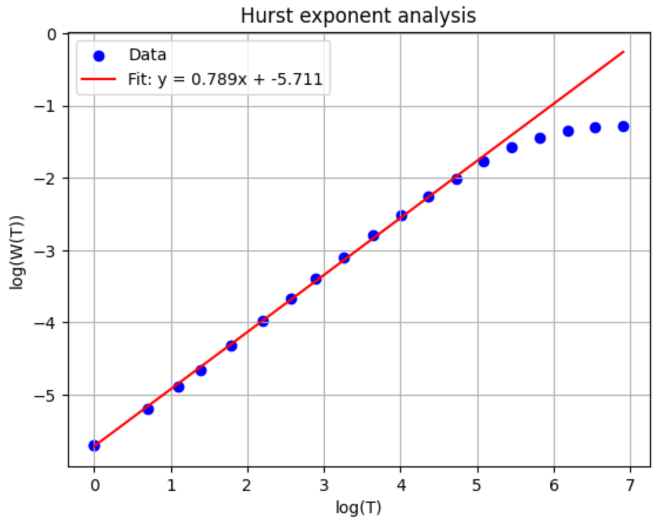

Figure 2: Log-log plot of the rescaled fluctuation function $W(T)$ versus window size T for a single product demand series under the parameters $N = 100$, $L = U = 50$ and $M = 250$. The blue marks show $log(W(T))$ computed over increasing T, and the red line displays the least-square fit, yielding a Hurst exponent estimate of $H \approx 0.789$

In our analysis, we look at the fluctuations in demand $D(t)$ similar to how price movements are studied in econophysics by examining the quantity $s_j(t) = \log(D(j, t))$ [6]. We consistently use natural log throughout this thesis. The change $\Delta s_j = s_j(t + T) - s_j(t)$ captures how demand shifts over a time interval T. To analyse the nature of these fluctuations, we compute the mean square displacement $W_j^2(T) = \langle(\Delta s_j(T))^2\rangle$, averaging over all starting points t, and then define $W_j(T) = \sqrt{W_j^2(T)}$. We expect

$$W_j(T) \propto T^{H_j}, \tag{4}$$

where H_j is the product specific Hurst exponent [6]. To find H_j we plot $\log(W_j(T))$ versus $\log(T)$ and estimate the slope, which gives us H_j, as seen in Figure 2. We sample $W_j(T)$ over a wide range of T to identify where the power-law scaling breaks off, but fit the regression only to the intermediate T values that exhibit clear $W_j(T) \propto T^H$ trend, thereby avoiding distortions from edge of sample effects. To reduce statistical uncertainty, we estimate the Hurst exponent for each of the L tradable products' demand series $D(j, t)$ and report the overall $H = \langle H_j \rangle_{j=1}^{L}$ as the average of those L estimates. This approach is based on the observation that the Hurst exponent remains nearly constant across products, as they do not differ qualitatively in any intrinsic way.

Theoretical analysis

With the goal of explaining how need- and greed-based interactions generate differing Hurst exponents, we first aim to understand the model's behaviour from a theoretical standpoint. In this chapter, we analyse the model's dynamics and attempt to reproduce key results through logical reasoning and simplified mathematical arguments. We begin by examining the two core components of the model - need- and greed-based trading - followed by an analysis of the more complex case where both are combined.

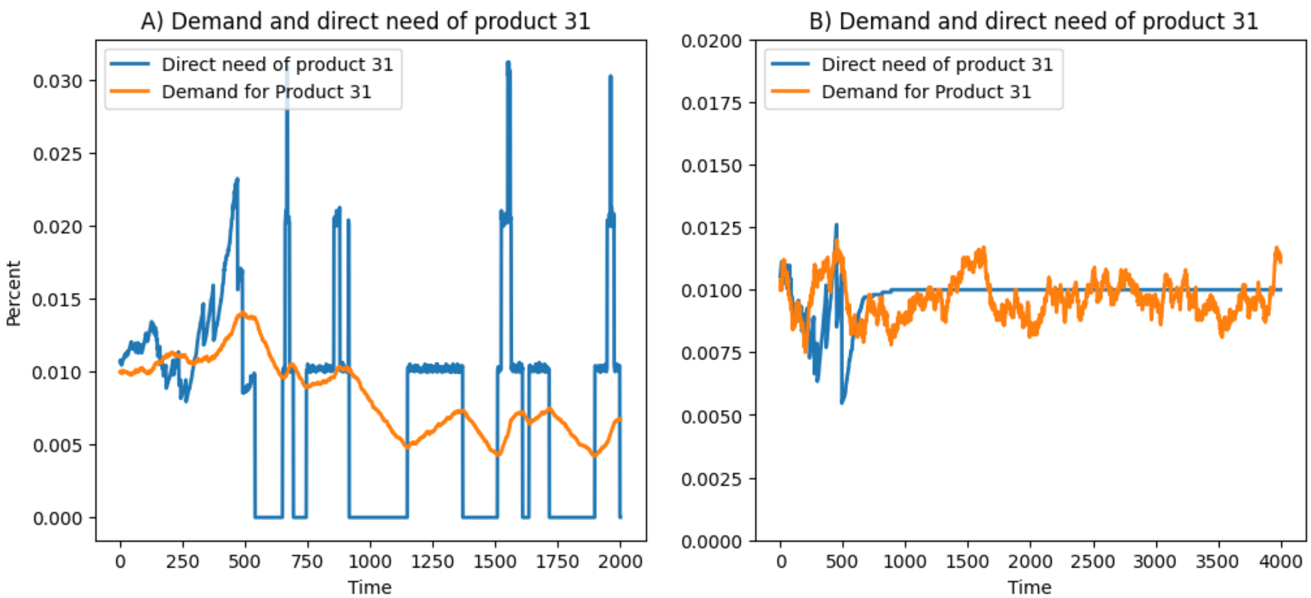

Figure 3: Evolution of demand and direct need for product 31 under two trading regimes. A) $N = 100$, $L = U = 100$, $M = 300$: both need-based and greed-based trades occur. B) Same parameters except $U = 99$: only need-based trading is possible. In both cases, demand closely tracks direct need, but in (B) the direct-need fraction rapidly plateaus at a constant level of L^{-1}.

Need trading

As discussed the model runs of an interplay between need and greed trading. When combining the measures of **direct need** $A(j)$ and **demand** $D(j)$ as in Figure 3, we observe a clear correlation: demand tends towards direct need. We will return to the analysis of this phenomenon, and discuss exactly why it happens later, but first important insight into the model is to gain from analysing the edge cases of pure need trading and pure greed trading separately.

To create a scenario where the agents only trade based on need, one simply needs to set $U < L$ as each agent in this regime will bee of need of at least one item at all times. When $U < L$, each agent may start with an inhomogeneous inventory as a consequence of the inventory initialisation process. However, since each agent certainly has needs, they are never able to request an item they already possess, and are therefore unable to obtain multiple copies of items beyond what they were initially given. Consequently, after a short transient period, all missing items are uniformly distributed, $a_j = N\frac{(L-U)}{L}$, as each agent ends up with one copy of U items and zero copies of the remaining $L - U$ items. As a result, direct need stabilises at a value of $A(j) = \frac{1}{L}$.

This state is considered an attractor, as regardless of the initial inventory configuration, the model converges toward this steady state of optimally balanced inventory [7]. With no fluctuations in direct need to drive changes in demand, the resulting demand graph for each good behaves like noise around $\frac{1}{L}$.

This is exactly what we observe in Figure 3B, where a transient period of approximately 1000 time steps is followed by a stable value of direct need at $A(31) = \frac{1}{L} = 0.01$, with demand fluctuating around it like noise.

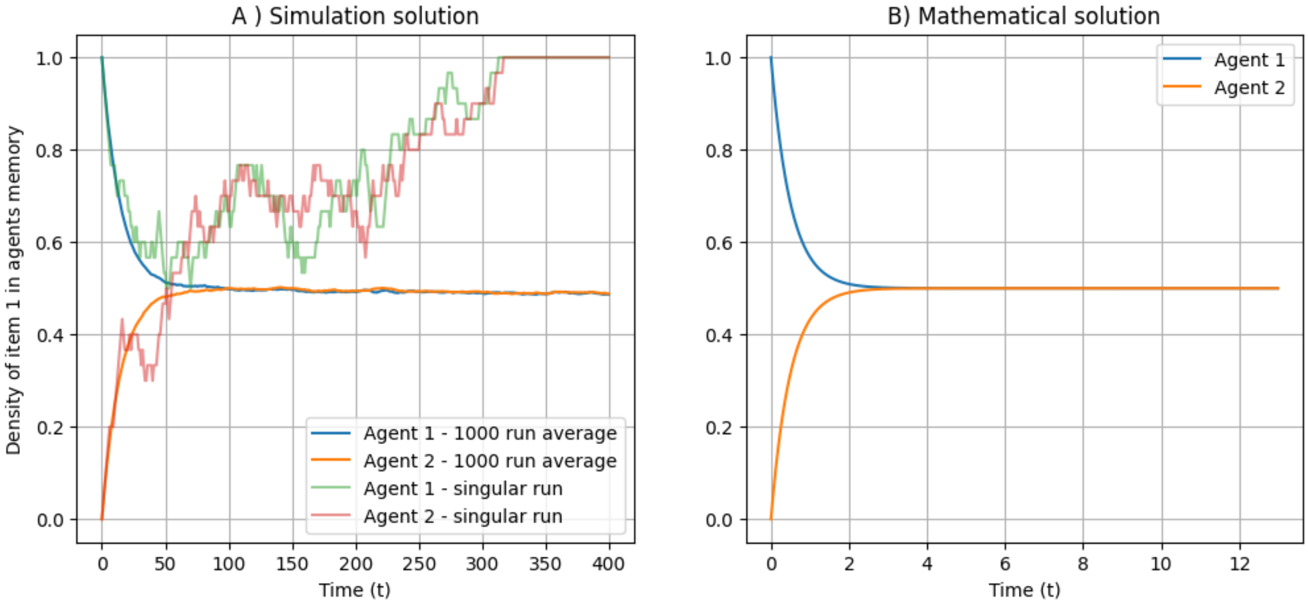

Figure 4: A) Simulation results for a simple two-item case ($L = 2$) with two agents ($N = 2$), interacting purely through greed-based trading, each with memory length $M = 30$. Initially, Agent 1's memory contains only item 1, while Agent 2's memory contains only item 0. The plot shows a single run where the agents' memories evolve toward an absorbing state, along with the average over 1000 independent runs. B) Numerical solution of the corresponding mathematical model from plot A), solved using a 4th-order Runge-Kutta method [8].

Greed trading

To thoroughly understand the mechanics behind greed trading we will consider the simplest case of two agents trading two products $N = L = 2$. Each agent has a distinct memory list of same length respectively called M_1 and M_2 as they also would in our model. As the system contains only two products, we chose that the products have indices 0 and 1 in the agents' memory. When trading, these indices result in each agent having

$$\rho_i = \frac{1}{M} \sum_{j=1}^{M} x_j \tag{5}$$

as the probability of choosing product 1, where x_j is the j-th element of the memory list, and probability $q_i = 1 - \rho_i$ of choosing product 0. When receiving a product, ρ_i and q_i also represents the chance of product 1 or 0 being replaced in the memory. This simplistic scenario of our only greed trading model is easily modelled mathematically by analysing the change in memory:

$$d\rho_1 = \rho_2 q_1 - \rho_1 q_2 \tag{6}$$

$$d\rho_2 = \rho_1 q_2 - \rho_2 q_1 \tag{7}$$

Showing that ρ only changes if a request for product 1 replaces product 0 in memory, or vice versa. The system converges towards the steady-state solution defined by $d\rho_i = 0$ [7]. From Equation 6 and 7 we see that the general condition for the steady-state is:

$$\rho_1 q_2 \;=\; \rho_2 q_1 \;\Rightarrow\; \rho_1(1-\rho_2) \;=\; \rho_2(1-\rho_1) \;\Rightarrow\; \rho_1 \;=\; \rho_2$$

A sanity check confirms that this makes sense, since antisymmetric updates cannot reduce the difference between two equal memory values.

Analysing Figure 4B shows that the ODE model drives both memories to the average of their initial values and then holds them constant. For example, if $\rho_1(0) = 1$ and $\rho_2(0) = 0$, both converge to $\rho = 0.5$. If instead $\rho_2(0) = 0.5$, they converge to

$$\rho \;=\; \frac{\rho_1(0) + \rho_2(0)}{2} \;=\; 0.75.$$

In practice (Figure 4A), a single run reaches the steady state $\rho_1 = \rho_2$ after a transient period, but finite-M noise eventually pushes both ρ_i to either 0 or 1. These two points constitute our models so called fixed points [7]. At these points both agents' memories are filled entirely with product 1 or entirely with product 0, i.e. $\rho_1 = \rho_2 = 1$ or $\rho_1 = \rho_2 = 0$, meaning that each agent can only request and receive the same product, effectively locking the system in a standstill. That is why a single run ends with one product dominating memory. However as no single product is inherently favoured, the 1,000-run average stays near 0.5, as seen in Figure 4.

Note that the evolution towards a fixed point is not a runaway feedback loop: if a product occupies 99% of memory, it has only a 1% chance of replacing a slot that does not already contain itself. Conversely, a 1%-share product has a 99% chance of expanding if selected. These opposing probabilities balance out, creating the random walk pattern that is observed.

In the more complex case of N agents and L items, we would expect our model to behave under the same principles: each agent's memory converges toward a state of an average of the systems initial total memory and then performs a collective random walk. However when multiple products exists simultaneously, what happens is not that one product will immediately dominate the memory, but instead that unpopular products slowly "die out", fading from the collective memory. This happens when an item occupies zero slots in any agent's memory, effectively leaving it forgotten, since no agent can request it again. Consequently, the collective memory should execute a random walk, progressively forgetting items and eventually converging toward a single product filling the entire collective memory, corresponding to $D(j) \;=\; 1$.

One reason we expect similar behaviour, is that an L product system can always be viewed as a sort of 2 product system, separated in product i and product "not i". Either product i will reach the dominant state, having eliminated all other products, or it will be forgotten, after which the system can be split again into product j and "not j", repeated until a single product is left. As for why N agents does not affect the model, it comes down to the fact that all trades are 1-on-1, and that the expected change in the two agents memories will still be opposite but equal as in equation 6 and 7.

Figure 5A shows the outcome of ranking products based on their total cumulative demand over a single simulation and then averaging across 1000 simulations. The figure displays how the average demand for each rank evolves over time. We clearly observe the phenomenon of products being forgotten. Interestingly, it appears that the time it takes for a product to be forgotten increases as fewer products remain. This prolonged survival is a direct consequence of the remaining products occupying increasingly large portions of memory. Initially, each product fills approximately $\frac{1}{L}$ of

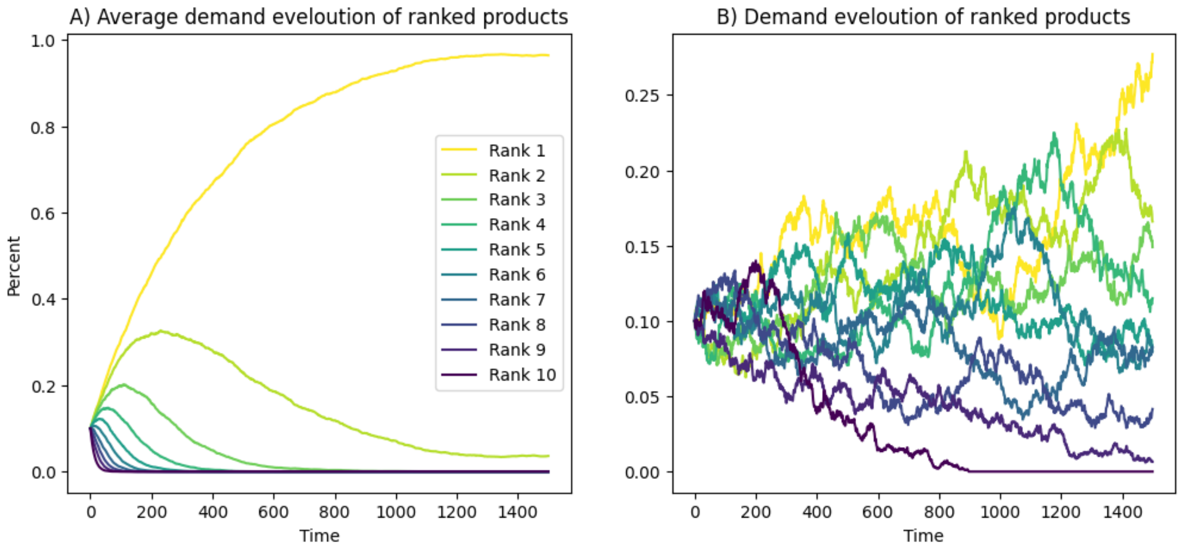

Figure 5: A) Average demand evolution over 1000 simulations, where products were ranked by total demand accumulated over the entire simulation duration. The results illustrate a strong battle royale effect, where only one item is allowed to prevail as a consequence of the system's fixed points. B) Temporal demand plot illustrating the behaviour of demand in a single simulation of our greed-only model. The legend shows products ranked from highest (Rank 1) to lowest (Rank 10) accumulated demand and applies to both Figure A and B. Model settings were $N = 100$, $L = 10$, $U = \infty$. In A), $M = 2$ was used for faster simulation time, and in B), $M = 20$ was used to better illustrate the dynamics.

the collective memory, but as products are forgotten, the total number of memory slots remains constant while the number of effective items in the system decreases. As a result, the probability of a product reaching an absorbing state - where the product is entirely forgotten - decreases as the number of remaining items declines. Measuring the Hurst exponent of the demand time series in the greed-only trading case yields $H \approx 0.5$, indicating that the demand observed in Figure 5B, and in general, behaves like a random walk if no need is present. On a run-to-run basis, there is no way to classify one product as inherently better than another, since the demand for each product undergoes a random walk until it reaches either $D(j) = 0$ or $D(j) = 1$. However, because these two fixed points are absorbing states in the model [7], we find that if the simulation is allowed to run for a sufficiently long time, one product will eventually dominate - just as observed in our simplified model. This dominance does not arise from any intrinsic value, but rather from pure randomness.

Combination of need and greed trading

Having understood the fundamental principles governing need-based and greed-based trading in isolation, we now turn to the more complex and realistic regime where both interact. This is important for our overall goal of understanding how value can emerge from agent interactions, as value emergence only becomes non-trivial when both forces interact. Our model has four main parameters - N, L, U, and M - which together allow for a myriad of ways to run and explore the system.

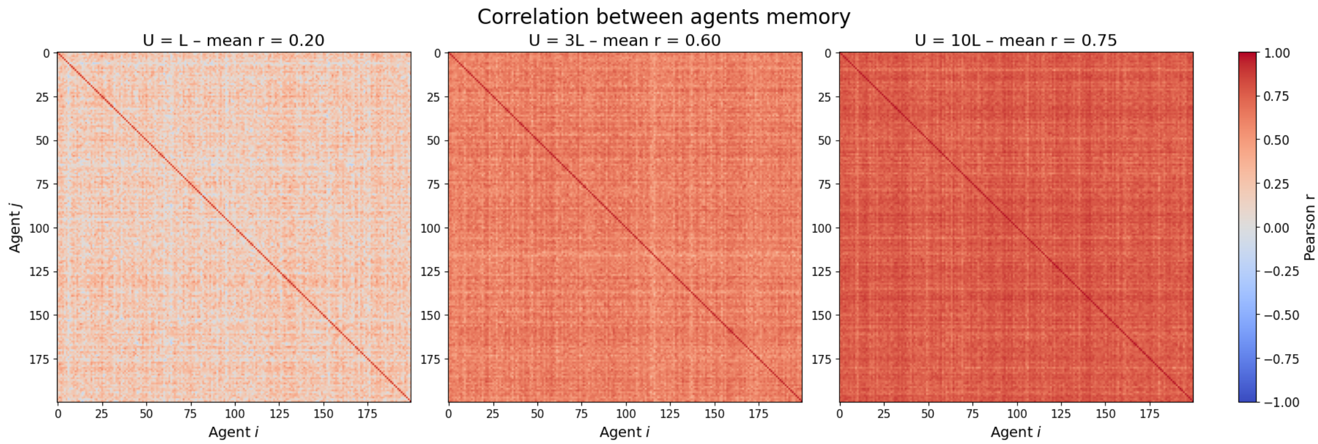

Figure 6: Pairwise correlation of agents' memory histograms at the end of the simulation, for increasing wealth parameter U. Each panel shows the Pearson correlation matrix of the final memory histogram - where each agent's memory of length $M = 300$ is summarised as fractions each $L = 100$ items take up of the memory - computed over $N = 200$ agents. From left to right, $U/L = 1,\ 3,\ 10$ (ie. $U = 100, 300, 1000$). The Pearson correlation rises from 0.20 to 0.60 to 0.75 [9]. When $U = L$, need-based trades dominate and agents' memories remain relatively idiosyncratic. As U/L rises, greed-based trading prevails, driving each agent's memory toward the population average and thus homogenising the collective memory distribution.

Boundary of only greed trades

When $U \gg L$, almost all trades are greed-driven, allowing us to analyse the model on the boundary to only greed trading. In greed mode, memories rapidly "synchronise" via cross-trades, causing the high pairwise correlation seen in Figure 6. Thus, the system effectively behaves like a pure-greed regime.

In Figure 6, we observe that agents' individual memories still tend to converge toward the collective memory, even in the more complex case of $N = 200$ agents and $L = 100$ products. This analysis is done on the basis of calculated Pearson correlation matrices [9], where the Pearson correlation coefficient r is defined as

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}. \tag{8}$$

Here x and y in our case represent two agents' memory, and r then takes a value between ± 1 indicating how correlated the two memories are with $r = 1$ being perfectly correlated [10, 9].

The correlation effect observed in Figure 6 appears to strengthen as the system becomes increasingly dominated by greed-based trading, evident by the rising correlation value r from 0.20 to 0.60 to 0.75 as U rises [9]. This suggests that in scenarios where U is close to L, agents behave less like a collective "hive mind" and more as independent decision-makers, each prioritising different products.

Boundary of only need trades

The other boundary regime occurs at $U = L$ and is dominated by need-based trading: an agent can only act on greed when they already hold at least one of every product. In this regime, the

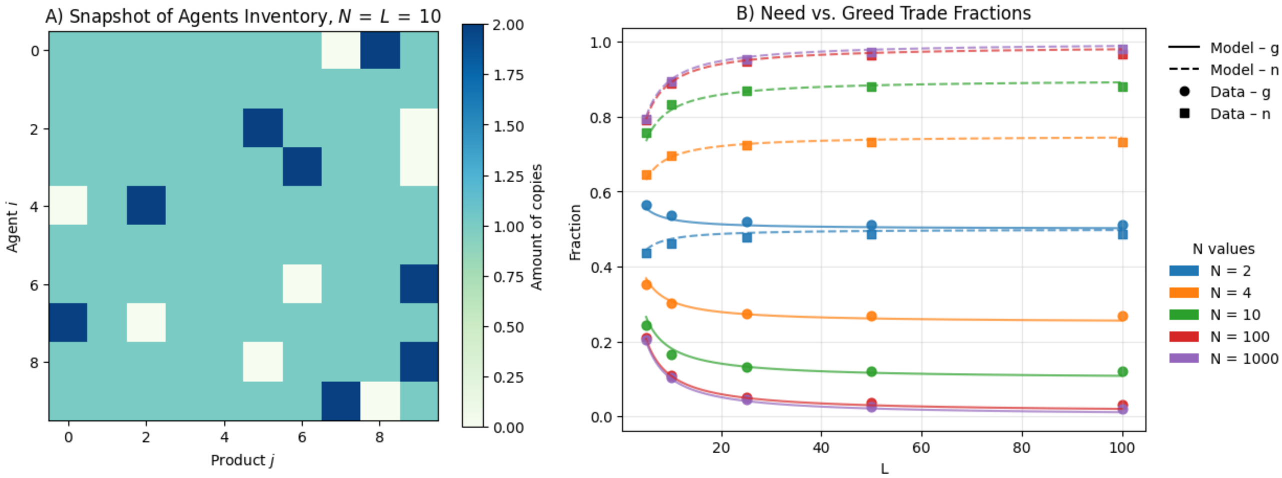

Figure 7: A) Heatmap snapshot of the inventory matrix for a system with $N = L = 10$. Each row corresponds to an agent and each column to a product; the colour indicates how many copies of that product the agent holds. The snapshot takes place after the short need-driven transient period. Here all agents occupy one of two states, need- or greed-state (missing exactly one item and having two copies of one other, or having exactly one copy of every product), permitting occasional greed trades only when a perfectly sorted inventory momentarily arises. B) Measured fractions of need-based (squares) and greed-based (circles) trades as functions of product diversity L for various agent counts N. Curves are shown for N = 2 (blue), 4 (orange), 10 (green), 100 (red) and 1000 (purple), with the dashed lines (n) and solid lines (g) marking the predicted fraction of need and greed trades by the mathematical models.

model no longer behaves as smoothly as it did close to the pure-greed boundary. We instead observe interwoven positive and negative feedback loops, reflected by a higher Hurst exponent in this regime ($H > 0.5$). To understand these feedback loops, we analyse when and how often greed and need trades occur. Since need-based trading is the fallback rule, the pure-need dynamic still applies: need trades can only eliminate existing shortages and never create new ones for the asking agent. Consequently, once agents' inventories have been optimally redistributed, the agents will as in the need case have one copy of U products, and zero copies of $L - U$ products. The trading model differs from the need-only case in that $L - U = 0$, which means all needs are satisfied allowing greed-trading to take place. As a result, agents can occupy only two distinct inventory states:

Greed: A fully sorted inventory, which leads to greed-trading.

Need: A almost-sorted inventory missing one product and holding two of another, which triggers need-trading.

These two inventory states are illustrated in Figure 7A: agents $i = 1, 5$ occupying state greed, while the rest occupy the need state. With two distinct inventory states, four types of trades can occur: a trade can shift an agent from greed to need, from need to greed, or leave the inventory state unchanged. Each trade can occur with its own probability provided in the Table 1.

The easiest probability to justify is the likelihood of remaining in a state of greed. An agent in this state holds one copy of each product and is about to give one away in a trade. The only way to maintain a fully sorted inventory is if the agent receives exactly the same product it gives away,

Table 1: Transition probabilities for the two inventory states "Greed" and "Need". Diagonal entries give the probability of remaining in the same state, while off-diagonal entries give the probability of switching, in terms of the parameters L and N.

	Greed	Need
Greed	$\frac{1}{L}$	$\frac{1}{L}+\frac{1}{N-1}-\frac{1}{L(N-1)}$
Need	$1-\frac{1}{L}$	$1-\frac{1}{L}-\frac{1}{(N-1)}+\frac{1}{L(N-1)}$

which occurs with probability $\frac{1}{L}$. Since one outcome must happen, the probability of transitioning from greed to need is simply the complement: $1-\frac{1}{L}$.

The transition from a need state to a greed state is more complex, as it can occur through two distinct scenarios. In the need state, an agent always requests the product it lacks. To achieve a fully sorted inventory, the trading partner must request the product of which the agent has two copies of, so the agent ends up with one of each product.

One way this can happen is if the trading partner randomly asks for that specific product, this happens with probability $P(A)=\frac{1}{L}$. However, since the product distribution is uniform throughout the system, if our agent has two copies of a product, there must exist another agent missing that product - who is by definition of need forced to request it. The probability of meeting this specific partner is $P(B)=\frac{1}{N-1}$ (i.e., one out of every other agent in the system).

Since these events are independent, we apply the inclusion-exclusion principle [11]:

$$P(\text{need}\rightarrow\text{greed})=P(A\cup B)=P(A)+P(B)-P(A)P(B)=\frac{1}{L}+\frac{1}{N-1}-\frac{1}{L(N-1)} \tag{9}$$

The probability of remaining in the need state is simply the complementary probability $1-P(\text{need}\rightarrow\text{greed})$, as shown in Table 1.

With the probabilities of each trade scenario defined, we can construct a matrix system to solve for the steady-state solution, yielding the constant fraction of greed trades g and need trades n.

$$\begin{pmatrix}\frac{1}{L} & \frac{1}{L}+\frac{1}{(N-1)}-\frac{1}{L(N-1)}\\ 1-\frac{1}{L} & 1-\frac{1}{L}-\frac{1}{(N-1)}+\frac{1}{L(N-1)}\end{pmatrix}\begin{pmatrix}g\\ n\end{pmatrix}=\begin{pmatrix}g\\ n\end{pmatrix}$$

Enforcing $g+n=1$ one can solve the system of equations and obtain:

$$g=\frac{N+L-2}{NL-1},\quad n=\frac{(N-1)(L-1)}{NL-1} \tag{10}$$

Comparing our mathematical model with simulation data across various values of N and L as seen in Figure 7B, we find that it predicts the system's behaviour almost perfectly, with a near negligible error. This close agreement between theory and simulation confirms that the simplified two-state model captures the essential dynamics of the $U=L$ regime, despite the underlying complexity of agent interactions.

Feedback loops in the need dominated regime

With an understanding of when greed trading occurs and how need-based trading influences demand, we can now uncover the mechanism behind the elevated Hurst exponent in this regime.

As discussed earlier, and observed in Figure 3A, demand closely tracks direct need. In a system governed purely by need-based trading, both quantities stabilise uniformly at $1/L$. However, in the $L = U$ system, direct need may vary resulting in some products being widely needed, while others are perfectly distributed across agents and thus not needed at all.

When a greed trade occurs, it almost always creates a "hole of need" for a specific product, effectively initiating a chain of need trades. This need for a specific product propagates from agent to agent until it is randomly fulfilled, with a probability given by Equation 9. As a result, demand for that product builds over approximately $\frac{1}{P(\text{need}\rightarrow\text{greed})}$ trades, directly observable in Figure 3A.

The resulting spike in demand means an increased probability that the product will be selected in subsequent greed-based trades as this probability is exactly equal to demand. The increased number of greed trades involving the specific product, in turn, further amplifies the number of need chains involving the same product. This forms a reinforcing positive feedback loop: need generates demand, and demand promotes further need.

At the same time, the reverse dynamic can also occur. As the need for a product decreases, fewer agents request it, leading to a gradual drop in demand. Furthermore, the system is net zero-sum, so any positive feedback that increases demand for one product must be accompanied by a corresponding reduction in the demand for other products. The reduction in demand decreases greed-based requests, and therefore also the amount of need generated for the product. Over time, this negative feedback can cause demand to fall below the baseline of $\frac{1}{L}$, as seen in Figure 3A.

In general, this positive feedback loop is strongest when demand is low, as need has a greater chance of raising demand in this regime. As previously discussed in the context of pure greed-based trading, a product occupying just 1% of memory ($D(j) = 0.01$) has only a 1% chance of being selected in a trade, but a 99% chance of increasing its presence in global memory if traded. Conversely, a product that already fills 99% of memory has just a 1% chance of further expanding. While these effects balance out under pure greed, the inclusion of need-based trading disrupts this balance and strengthens the positive feedback for low-demand products.

The positive feedback mechanism is inherently constrained. In theory, no more than half the agents can simultaneously need the same product, since each agent lacking a product must be balanced by another agent holding two copies of it. As demand increases, it becomes harder to generate new need chains for that product, simply because the large number of agents already lacking it creates a surplus of product among the remaining agents. Additionally, as demand for a product increases, trades begin to fail more frequently - agents who need the product have no copies to offer when encountering others, whether those are in need or making greed-based requests. As in the pure-greed regime, rising demand alone cannot sustain further demand growth. Combined with the reduced creation of need, this accelerates the product's redistribution throughout the system, suppressing additional demand and thus providing an additional negative feedback when $D(j)$ is large. This saturation effect prevents the system from spiralling into a state where one product monopolises 100% of the demand.

In addition to positive and negative feedback loops, other mechanisms - such as the fading of products from memory - also contribute to the elevated Hurst exponent. When no agents need a particular product, it gradually disappears from the collective memory, causing its demand to approach zero. However, if the memory size M is sufficiently large relative to the amount of products L, it is rare for a product to reach $D(j) = 0$. As explained above, the positive feedback loop between direct need and demand weakens when demand is high and, conversely, strengthens when demand is low - allowing products to recover even after nearly being forgotten.

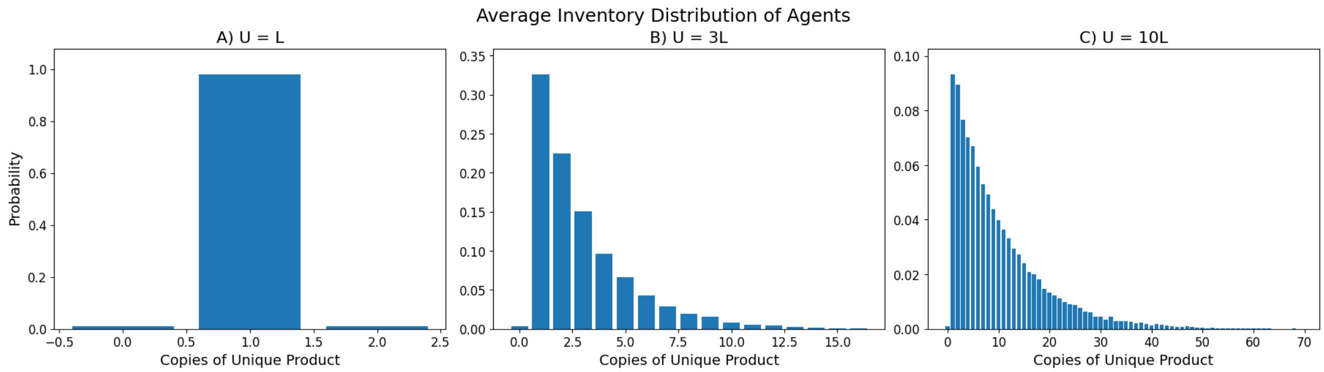

Figure 8: Average inventory distribution of agents after the transient, for increasing wealth parameter U. Each panel shows the probability that an agent holds k copies of a given unique product - computed by taking the inventory histogram and normalizing the mean bin-count over $L = 100$ products across $N = 200$ agents. From left to right, $U/L = 1$, 3, 10 (i.e. $U = 100, 300, 1000$). The distribution shifts from a near-delta spike at one copy to increasingly broad, exponential-like tails. When $U = L$, need-based trades enforce exactly one copy per agent; as U/L rises, greed-driven exchanges permit accumulation of multiple copies, yielding heavy-tailed inventory profiles.

Extended regime: Surplus-inventory regime

When U exceeds L, agents' inventories can take on many more configurations than in the $U = L$ case, where only two states existed. With $U = L + 1$, an agent in the "greed" state still has one copy of every product plus one extra of a random type - altering the chance of staying in or leaving greed. The "need" state also splits into multiple variants: an agent might be short of one product while having two copies of two different products, or three copies of one. Each variant has its own transition probabilities back to need or into greed. In Figure 8 we collected each agent's inventory after the transient, counted how many copies k each agent i had of product j, then repeated for all j and i, and normalised to obtain $P_i(k)$, the probability that an agent i holds k copies of a given unique product. We then average across i to get a single distribution, as shown in Figure 8. Increasing U clearly adds a layers of complexity, leading to different effects on the systems dynamics.

The most prominent effect is a natural increase in the fraction of greed trades. As overall wealth rises, needs become less frequent, reducing the number of need-based trades.

When U exceeds L, the theoretical maximum fraction of agents in need rises - approaching $N - 1$ as the possible amount of "hoarding" $\epsilon = U - L$ grows - allowing rarer but more extreme need spikes as seen in Figure 9. For $U = 5L$, need can momentarily reach 0.4 (twenty times its low baseline of 0.02), whereas for $U = L$ the baseline is around 0.08, so a 0.4 peak represents only a fivefold increase. In Figure 9, the $U = L$ demand curve shows low-amplitude, high-frequency demand fluctuations consistent with trend lengths of $T \approx \frac{1}{P(\text{need}\to\text{greed})}$. By contrast, $U = 5L$ exhibits larger-amplitude shifts that persist well beyond $\frac{1}{P(\text{need}\to\text{greed})}$.

In a purely greed-based regime, trades appear largely random, which could explain the larger amplitude shifts. However, Hurst-exponent analysis (discussed later) reveals trends that persist well beyond $T \approx \frac{1}{P(\text{need}\to\text{greed})}$. This suggests that these extended feedback loops are genuine - and not merely random noise - indicating more complex dynamics than in the $L = U$ case.

Finally, larger U introduces a longer transient period. The initial distribution of product copies is random, producing a Gaussian distribution for the number of copies each agent holds of each

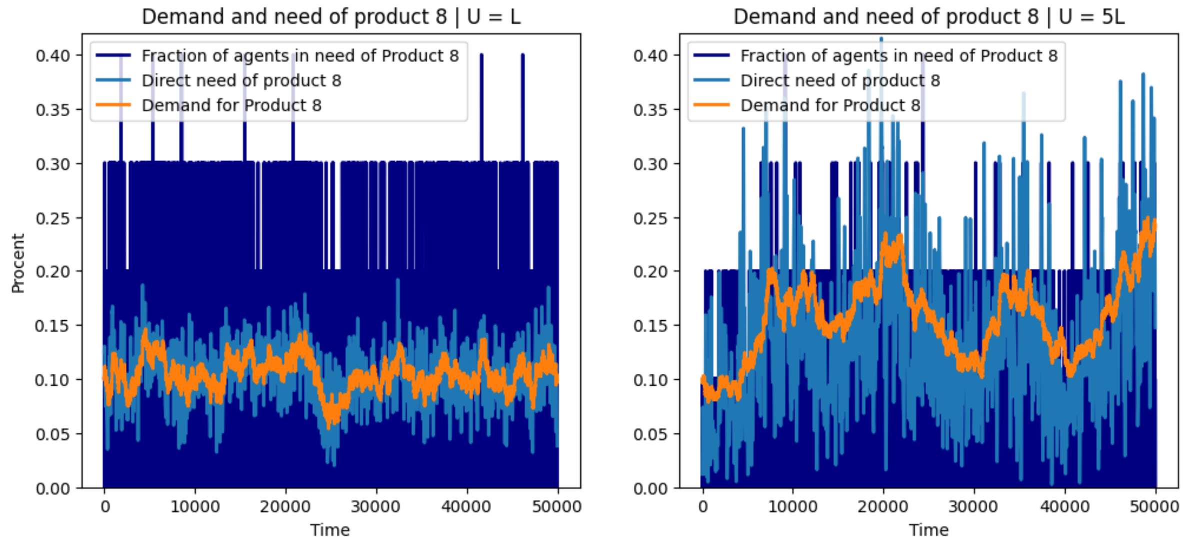

Figure 9: Temporal behaviour of product 8's need and demand for $U = L$ (left) versus $U = 5L$ (right). Each panel plots over time the fraction of agents in need of product 8 (dark blue), the temporally smoothed direct need (light blue), and the total demand (orange), for $N = 200$, $L = 10$. When $U = L$, at most $N/2$ agents can be simultaneously in need (baseline ≈ 0.08, peak ≈ 0.40, a ~ 5 times deviation); for $U = 5L$, the cap rises toward $N - 1$ (baseline ≈ 0.02, peak ≈ 0.40, a ~ 20 times deviation). Higher U yields rarer but more extreme bursts of need and demand, with larger amplitude, variance, and longer trend persistence.

product. This contrasts sharply with the heavily left-skewed distribution shown in Figure 8, where agents predominantly hold one copy of each product, and the probability of holding additional copies decreases steadily. Consequently, when U is large, transforming the initial Gaussian distribution into the final skewed distribution takes more time, resulting in an extended transient period.

Given the more complex nature of the $U > L$ regime, a relatively complete explanation - such as the one provided for the $U = L$ case - has not yet been established. In principle, one could derive a general master equation to track $P(k)$ for all k and U values, but calculating the corresponding steady-state fractions g and n entails its own computational challenges. Therefore, we restrict ourselves to describing the phenomena that emerge, leaving a thorough analytic treatment for future work. Even without a complete solution, the model behaves largely as expected and continues to exhibit the same positive and negative feedback loops. These dynamics help drive the system toward cooperative emergence of value - as seen in the Donangelo-Sneppen model [3] - even if the exact mechanisms become harder to isolate. The heavy dependency on greed-based trading introduces more variability and longer transients, making the system less analytically tractable, but still qualitatively consistent with the underlying trading logic.

Hurst exponent analysis

The Hurst exponent provides a way to compare our model to real-world markets. As presented earlier, real markets typically exhibit Hurst exponents in the interval $[0.55, 0.65]$ [6]. In our model, we identified three distinct trading modes - greed-only, need-only, and a mixture of the two - each leading to a different Hurst exponent.

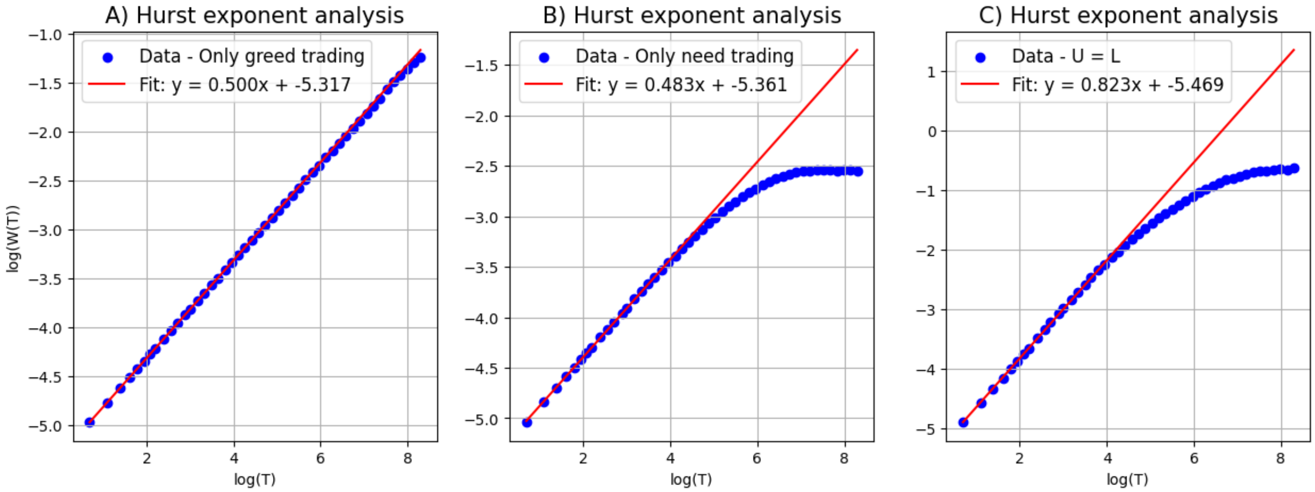

Figure 10: Fluctuation analysis of demand time series for three trading regimes. Each panel plots $\log W(T)$ versus $\log T$ for a linear fit (red line) whose slope is the Hurst exponent H. All simulations use $N = 50$, $L = 50$, $M = 300$. A) Greed-only trading: $H = 0.500$, consistent with random-walk fluctuations. B) Need-only trading ($U = 40$): $H = 0.483$, likewise near the mean-reverting regime. C) Combined need and greed trading ($U = L = 50$): $H = 0.823$, revealing strong persistence.

As shown in Figure 10, these three modes correspond closely to the three regimes of Hurst exponents discussed in Section *Hurst exponent* ($H < 0.5$, $H = 0.5$, $H > 0.5$).

Greed-only trading (Figure 10A) results in a Hurst exponent of $H = 0.500$, as discussed earlier. This corresponds to a pure random-walk regime. Since need does not exist in this mode of trade, all exchanges are driven solely by demand. As we have shown, demand-based decisions in isolation are essentially random, leading to a system-wide random walk in the collective demand.

Need-only trading (Figure 10B) results in $H = 0.483$, indicating that the demand time series for each product is mean-reverting. This aligns with expectations, as we previously discussed how direct need stabilizes at a constant value of $A(j) = \frac{1}{L}$. As a result, demand fluctuates around this baseline, behaving like noise that reverts back toward L^{-1} over time, resulting in mean-reverting behaviour for $s_j(t) = \log D(j,t)$.

Need- and greed-based trading at $U = L$ (Figure 10C) results in a Hurst exponent of $H = 0.823$. This is a very high Hurst exponent, even higher than what empiric data measures for real markets[6]. In this regime of Hurst exponents, our demand time series exhibits memory of earlier movement. This means that our demand tends to continue in the same direction. Theoretical thinking lead us to the conclusion that need-based trading chains of the length $\frac{1}{P(\text{need}\to\text{greed})}$ lead to this persistence in demand movement, together with other feedback loops.

In Figure 11A, we observe how demand behaves in intermediate cases of our model, where U ranges above L. In these scenarios, the Hurst exponent falls between the values observed for $U = L$ and pure greed trading. Another interesting observation can be made from Figure 11B, which shows $\log(W(T))$ at different values of L. As L increases, the demand series tends to follow the relationship described by Equation 4 over larger time scales T - notably in a way where $\log(W(T))$ begins to deviate when $T \approx \frac{1}{P(\text{need}\to\text{greed})}$. This suggests that the positive and negative feedback loops identified earlier are the primary drivers behind the elevated Hurst exponents in the $U = L$ regime. However, the Hurst exponent remains above 0.5 well beyond the $T \approx \frac{1}{P(\text{need}\to\text{greed})}$

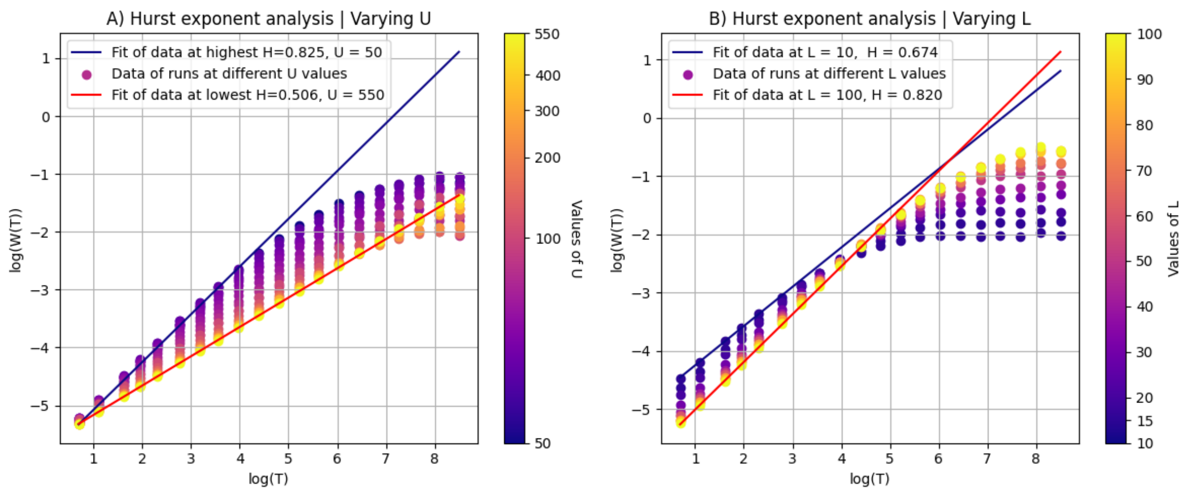

Figure 11: Hurst exponent analysis of varying values of the parameters U and L. In both figures, $N = 100$. A) $\log(W(T))$ is plotted for different values of U, with the number of distinct products fixed at $L = 50$ and memory size $M = 300$. The fits for the data corresponding to the lowest and highest Hurst exponents are also shown, occurring at $U = 50$ ($H = 0.825$) and $U = 550$ ($H = 0.506$), respectively. B) $\log(W(T))$ is plotted for different values of L, with wealth set relatively as $U = L$. To ensure consistent model behaviour across simulations, the memory size was scaled with the number of products as $M = 5L$. Trends break off around $T \approx \frac{1}{P(\text{need}\to\text{greed})}$, visible as a flattening in log-log plot. Fits are also shown for the cases with the lowest ($L = 10$) and highest ($L = 100$) product counts.

cut-off, suggesting that need-chains may be a key driver of the elevated Hurst exponent - though not the only one, as positive and negative feedback loops persist far beyond the specific case of $U = L$, as shown in Figure 11.

It is worth noting that the memory length M needs to be sufficiently large - typically around $M \approx 5L$ - to allow for elevated Hurst exponents. Shorter memories tend to allow more products to be forgotten as discussed earlier, this leads to a lower effective value of L and thus also a lower hurst exponent as seen in Figure 11B. Low values of M also limit positive and negative feedback loops that drive persistent demand dynamics. Aside from this lower bound, however, changes in M don't significantly affect the overall behaviour of demand, meaning there's a kind of sweet spot where memory is just long enough to capture trends without requiring too much computational power. These observations were also made in R. Donangelo and K. Sneppens paper *Cooperativity in a trading model with memory and production* [3], and are illustrated in Figure 13B.

To reproduce a Hurst exponent similar to that observed in real markets, it is clear that our model must operate in an intermediate regime where $U > L$. Choosing $U = 2L$ yields a Hurst exponent close to 0.6 for larger values of L and introduces an element of mystery, as the data continues to follow the proportional relationship described by Equation 4 over longer time scales T than $\frac{1}{P(\text{need}\to\text{greed})}$. This gives rise to interesting persistent trends, such as those seen in Figure 9 for the case $U = 5L$. It also reflects the nature of real markets, which are driven by a combination of partially predictable factors and underlying randomness that ultimately makes them difficult to forecast.

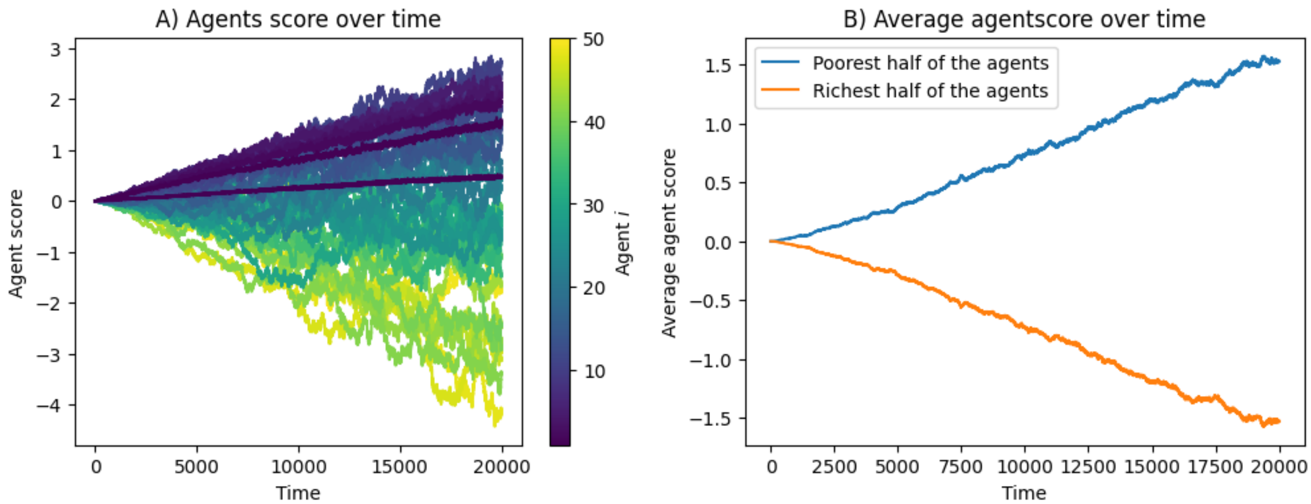

Figure 12: Comparison of agent performance across different wealth levels, represented by varying U_i. The system parameters are $N = 50$, $L = 25$, $M = 125$, with $U_i = 25 + 10i$, where i ranges from 0 to 49. A) Temporal evolution of individual agent scores. A clear inverse trend is observed, where agents with higher initial wealth tend to perform worse over time. B) Average performance of the richest 50% of agents compared to the poorest 50%. The poorer half consistently outperforms the richer half throughout the simulation.

Exploration of the model

Having carefully developed and explored our agent-based trading model, we are now in a position to use it as a tool for investigating real-world economic scenarios. The goal is not to exactly reproduce specific economic outcomes, but rather to explore how value emerges and evolves under different conditions - such as systems where agents' resources are unevenly distributed and where memory capacity varies. By testing these variations, we can assess the robustness of our findings and gain insight into how factors like wealth inequality or limited information shape the formation of value.

Unequal wealth distributions

To introduce unequal wealth among agents, we allow U to vary on an agent-by-agent basis by defining an agent-specific parameter U_i. Based on our earlier findings, we use the system configuration $N = 50$, $L = 25$, $U = 50$, and $M = 125$ as our baseline, as it yielded both Hurst exponents and demand behaviour most similar to those observed in real markets, while also keeping computational times low. Since small variations in these parameters do not significantly alter the model's overall behaviour, we do not strictly enforce that the total number of product copies, $\sum_{i=1}^{N} U_i$, remains constant across simulation runs.

With the parameters set, we tested several scenarios of unequal wealth distribution, including cases where a small fraction of the population holds most of the wealth, where the majority of agents are wealthy, and where wealth is evenly split between rich and poor agents. Across all these scenarios, the outcome remains consistent: agents classified as poor tend to outperform their wealthier counterparts according to our demand-based scoring system from Equation 3. Figure 12 illustrates this effect in a general case with a continuously varying wealth distribution, further confirming the robustness of this result. This outcome is not dependent on specific relative values of U_i and has been consistently observed across a wide range of wealth distributions. Agents with lower values of U_i engage more frequently in need-based trading and, as a result, tend to outperform agents who primarily rely on greed-based strategies.

From a theoretical viewpoint of our model, this makes sense, as we found demand to be largely driven by need-based trading in earlier chapters. An agent engaging in need trading has a statistically higher chance of asking for a product that is currently popular, as need is what creates value. Agents, on the other hand, who trade based on greed, theoretically act according to their perception of the market. However, as shown, agents are generally not good at forming an accurate picture of the market with need present, and it is ultimately safer to base trading decisions on their own needs. It is worth noting, however, that as seen in Figure 12A, a small level of wealth above $U = L$ appears beneficial, indicating that greed-based trading is not entirely disadvantageous.

Differing memory capacities

Throughout this paper, the effect of memory has rarely been discussed, despite its significance to the model. Using the same parameters and methods as in the analysis of systems with unequal wealth distributions, we also conducted an analysis of varying memory lengths among agents, transitioning from a global M to agent-specific M_i. Interestingly, no impact on agent performance was observed when varying memory length from $M_i = 25$ to $M_i = 500$ in the parameter regime we consider most similar to real markets. This suggests that memory length has little to no effect on an agent's predictive power of the market, and thus offers no clear real-world interpretation.

However, effects do appear that allow agents with shorter memories to perform better in markets almost purely driven by greed-based trading - that is, for large values of U. Figure 13A illustrates these findings, but the findings mainly reflect behaviour in the regime of pure greed trading and are not relevant to the dynamics of need-based trading, where demand follows more structured patterns. In this greed-dominated regime, demand behaves like a pure random walk, so performance differences are unlikely to be driven by meaningful trading patterns. Instead, the observed effects likely stem from how the system's memory converges toward the mean of the collective memory - though this remains speculative.

Outlook

Throughout this thesis we have analysed our variation of the Donangelo-Sneppen model more or less in isolation. Looking forward it would be interesting to observe how our findings align with real world markets, comparing it with concrete data sets. However, already now we can see that certain characteristics of the model align with popular economic theory.

Through our analysis of pure greed- and need-based trading, we found that value - measured in terms of demand - closely follows direct need. Moreover, when the system was deprived of any need, demand (and thus value) became entirely random and meaningless. This aligns with Menger's claim that value arises from a good's ability to satisfy human needs, rather than from any inherent property [2].

The subjective theory of value was evident not only in the pure regimes of need and greed, but also when we combined the two to replicate the Hurst exponent observed in real markets. Interestingly, agents considered poorer consistently outperformed their wealthier counterparts.

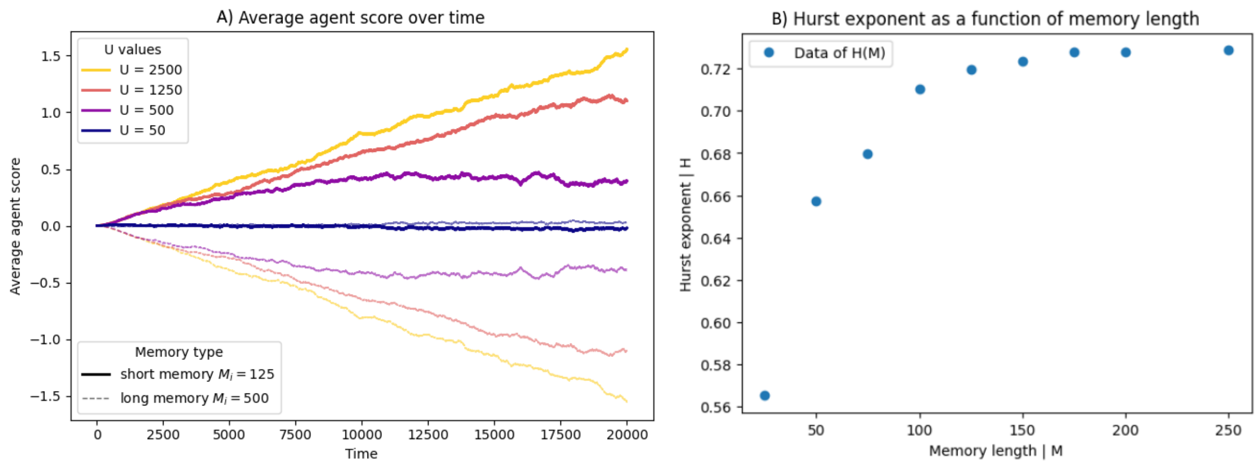

Figure 13: A) Evolution of the average agent score over time for four different wealth levels (U = 50, 500, 1250, 2500). The system parameters are $N = 50$, $L = 25$. Solid lines show agents with short memory ($M_i = 125$) and dashed lines their long-memory counterparts ($M_i = 500$). Higher U leads to progressively larger positive gains for short-memory agents and symmetrical losses for long-memory agents. B) Analysis of varying hurst exponent as a function of memory length. The system parameters are $N = 50$, $L = 25$, $U = 25$. Hurst exponent grows as a function of memory length, but plateaus at a maximum values as $M \approx 5L$.

Need emerges as the driving force behind value - supporting Menger's argument that goods acquire value only because they are perceived as means to satisfy specific wants. Agents engaging in need-based trading are not merely seeking valuable products; they are, in effect, creating them.

While it is somewhat disappointing that agents with better memory did not perform better, this outcome highlights a limitation of the model: it may lack the structural complexity or incentive mechanisms found in real markets, where informational advantage can lead to improved performance. Acknowledging this limitation is important, and it suggests a direction for future work - potentially integrating more sophisticated memory use or decision-making processes into agent behaviour.

Conclusion

In this thesis we explore a variation of the Donagello-Sneppen model [4]. We first investigate the internal dynamics of this model, before extending the model to more real-world inspired economic scenarios, such as cases with variable richness, or differing agent memory.

We have analysed the model's edge cases of pure need- and pure greed-based trading from an analytical standpoint. In the case of pure need trading, we found that the direct need for each product settles at a constant value of $\frac{1}{L}$ after a short transient phase. As demand closely follows direct need, it fluctuated around this constant, showing little capacity for meaningful evolution in the demand pattern. For pure greed based trading scenarios, we found that all agents converge into a collective hive mind with almost identical memories and trading patterns, slowly forgetting products until one product dominates all memories.

Analysing the intermediate regime, where both need- and greed-based trades occur, we found that product demand was primarily driven by need-based trading, exhibiting strong positive and negative feedback loops. In the regime closer to pure need trading, demand patterns were largely explained by observable need chains of length $T \approx \frac{1}{P(\text{need}\rightarrow\text{greed})}$. In contrast, as greed-based trading became more dominant, demand patterns were better described by the dynamics of these feedback loops.

To reinforce our findings and relate the model to real-world markets, we analysed the Hurst exponents across different trading regimes. In the pure greed regime, we found $H \approx 0.5$, supporting our theory that demand in this case behaves like a random walk. In the pure need regime, a value of $H \approx 0.48$ aligned with the observation that demand remained stable and fluctuated around direct need, resembling noise. When combining need- and greed-based trading, Hurst exponents ranged from $H \approx 0.82$ in strongly need-driven regions to $H \approx 0.5$ in greed-dominated ones. These higher Hurst values indicate persistent demand behaviour, consistent with our explanation of demand movement as driven by reinforcing feedback loops.

The model's need-driven value closely mirrors the early subjective value theory adopted by Menger and Hayek. In our model value arises from agents' direct need - supporting Menger's idea that worth stems from satisfying wants [2] - and emerges through decentralised interactions of memory and inventory, reflecting Hayek's view that complex market order can develop without central coordination [5].

Our model's simplicity comes at the cost of analytical tractability: even with minimal rules, its dynamics become too complex to solve exactly for the steady-state fractions of need (n) and greed (g) when $U \neq L$. Developing a general equation for g and n across arbitrary U would clarify the feedback loops driving demand in those regimes. Additionally, improving agents' memory - by, for example, weighting recent requests more heavily or introducing forgetting mechanisms - could reveal how better information retention impacts performance, bringing the model closer to real-world insights about the benefits (or drawbacks) of market knowledge.

Overall, our results highlight that - at least in a minimal barter-memory setting - subjective value indeed emerges from direct need, and that mixing even a small fraction of greedy, memory-driven trades yields the type of persistence typical of real markets. Deeper analytic treatment or changes to the core simulation may bring this model closer to economic reality.

Acknowledgement

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